

A Project Report On

Electricity Demand Forecasting & Peak Risk Detection System

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**This is to certify that the project work entitled
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submitted in partial fulfillment of the requirement for
the award of the degree of**

Master of Science (Data Science)

of the

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DECLARATION

We hereby declare that this project work entitled "**Electricity Demand Forecasting & Peak Risk Detection System**" is a record done by us.

We also declare that the matter embodied in this project is genuine work done by us and has not been submitted whether to this University or to any other University / Institute for the fulfillment of the requirement of any course of study.

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ABSTRACT

Accurate electricity demand forecasting and real-time peak risk detection are critical for modern power grid management, enabling utilities to balance supply and demand efficiently while minimising costs and preventing blackouts. This project presents an end-to-end machine learning system that integrates an XGBoost-based regression model for hourly electricity load forecasting with a gradient-boosted classification model for binary peak-hour detection.

The dataset spans two years (January 2022 – December 2023) of hourly records encompassing 17,496 observations with 19 engineered features including temporal indicators (hour, day-of-week, month, season), meteorological variables (temperature, humidity, solar irradiation), socio-economic indicators (population, GNI per capita, electricity price), and lag/rolling-window statistics (1-hour lag, 1-day lag, 7-day moving average).

The forecasting model achieves a Mean Absolute Error (MAE) of 1.84 MW, Root Mean Square Error (RMSE) of 2.38 MW, and an R^2 of 0.9964, indicating near-perfect fit with residuals below 2 MW. The peak detection classifier attains a Precision of 0.889, a Recall of 1.0, and an F1-Score of 0.941, backed by an AUC-ROC of 0.999, demonstrating excellent discrimination of peak demand periods that account for roughly 10 % of all hours.

A Streamlit web application provides an interactive dashboard with real-time 24-hour demand forecasts, peak-risk probability timelines, error distribution visualisations, and monthly aggregation trends — making the system accessible to grid operators without data science expertise.

CONTENTS

Chapter	Particulars	Page No.
1	Introduction	7
	1.1 Objective of the System	7
	1.2 Problem Definition	7
	1.3 Core Components	7
	1.4 Project Profile	8
	1.5 Assumptions and Constraints	8
	1.6 Advantages and Limitations	8
2	Requirement Determination & Analysis	9
	2.1 Requirement Determination	9
	2.2 Targeted Users	9
	2.3 Tool Details	9
	2.4 Library Description	10
3	System Design	11
	3.1 System Flowchart	11
	3.2 Dataset Design	12
	3.3 Preprocessing Steps	12
4	Development	13
	4.1 Model Details & Source Code Overview	13
	4.2 Application UI Design	13
	4.3 Test Reports	14
5	Proposed Enhancements	15
6	Conclusion	16
7	Bibliography	17

TABLE INDEX

Sr. No.	Table No.	Particulars	Page No.
1	Table 1.1	Requirement Determination	10
2	Table 1.2	Tool Details	10
3	Table 2	Dataset Design	13
4	Table 3	Test Reports	15

CHAPTER 1: INTRODUCTION

Electricity is the backbone of modern civilization, powering homes, industries, hospitals, and transportation systems. Efficient management of power grids requires precise knowledge of future electricity demand, often called load forecasting. When demand spikes unexpectedly — during summer heat waves or industrial booms — power systems risk blackouts, equipment failure, and economic losses. Conversely, over-procurement leads to wasteful generation and higher costs.

This project, "Electricity Demand Forecasting & Peak Risk Detection System", addresses these challenges by building a dual-purpose machine-learning pipeline: (1) an XGBoost regression model that predicts hourly electricity load (in MW), and (2) a gradient-boosted classification model that flags each hour as a peak or non-peak risk period. The models are served via a Streamlit web application that provides live dashboards, 24-hour ahead forecasts, and actionable peak alerts.

1.1 Objective of the System

- Forecast hourly electricity demand with sub-2 MW error.
- Detect peak-risk hours in advance with high recall (zero missed peaks is the target).
- Provide an interactive dashboard accessible to non-technical grid operators.
- Enable monthly and seasonal demand trend analysis.
- Support 24-hour ahead load scheduling decisions.

1.2 Problem Definition

Traditional statistical methods (e.g., ARIMA) are unable to capture the non-linear, multi-variate relationships driving electricity demand — particularly the interaction of weather, time-of-day, holidays, and economic conditions. Existing utility tools often lack real-time peak risk scoring and require specialised data engineering teams. This project fills that gap by delivering an automated ML pipeline with a user-friendly front-end.

1.3 Core Components

- Data Ingestion & Cleaning: Hourly power system dataset (2022–2023, 17,496 records).
- Feature Engineering: Temporal, meteorological, socio-economic, and lag features.
- Regression Model: XGBoost Regressor for continuous MW prediction.
- Classification Model: Gradient-Boosted Classifier for binary peak detection.
- Evaluation Module: MAE, RMSE, MAPE, R^2 , Precision, Recall, F1, AUC-ROC.
- Streamlit Dashboard: 4-page interactive web application.
- Pickle Store: Serialised model outputs (all_results.pkl, predictions.pkl, peak_analysis.pkl).

1.4 Project Profile

Project Title	Electricity Demand Forecasting & Peak Risk Detection System
Domain	Data Science / Energy Analytics / Machine Learning
Platform	Python 3.10+, Jupyter Notebook, Streamlit
Duration	Academic Year 2025–26 (Even Semester, Semester 2)
Team Size	Up to 3 students
University	Marwadi University, Rajkot

1.5 Assumptions and Constraints

- Dataset quality: The cleaned CSV is assumed to be free of structural errors after preprocessing.
- Stationarity: Feature distributions in 2022–2023 are assumed representative of future periods.
- Environment: The application requires Python 3.10+, Streamlit, and the pickle files in the same directory.
- Real-time integration: The current version uses historical forecasts; live SCADA integration is a future scope.

1.6 Advantages and Limitations

Advantages:

- R^2 of 0.9964 — near-perfect load prediction accuracy.
- 100 % Recall — no peak hour is missed by the classifier.
- Fast inference from pre-trained pickled models (< 1 s per query).
- Technology-agnostic front-end (runs on any machine with a browser).

Limitations:

- Model trained only on two years (2022–2023); may not capture longer-term trends.
- No real-time data feed integration — forecasts are based on historical feature patterns.
- Weather forecasts used as inputs must be provided separately by the operator.

CHAPTER 2: REQUIREMENT DETERMINATION & ANALYSIS

2.1 Requirement Determination

The requirements were elicited from two sources: (a) a review of power-grid load-forecasting literature showing the need for sub-5 MW hourly accuracy, and (b) an analysis of the provided power_system_load_cleaned.csv dataset. The following table summarises hardware and software needs.

Category	Specification	Minimum	Recommended
Hardware	RAM	8 GB	16 GB
Hardware	CPU	4-core	8-core
Hardware	Storage	5 GB free	20 GB SSD
Software	Operating System	Any (Win/Mac/Linux)	Windows 11
Software	Python	3.9	3.11
Software	Streamlit	1.28	latest
Software	XGBoost	1.7	2.0
Software	Pandas / NumPy / Plotly	latest	latest

2.2 Targeted Users

- Grid Operators & Energy Managers — who need 24-hour ahead demand schedules.
- Utility Companies — for capacity planning and generation dispatch.
- Data Science / Analytics Teams — to evaluate model performance and retrain.
- Academic Researchers — studying demand-side management and ML in energy.

2.3 Tool Details

Tool / Platform	Version	Purpose
Python	3.11	Core programming language
Jupyter Notebook	7.x	Model development & experimentation
Streamlit	1.32	Interactive web dashboard
XGBoost	2.0	Gradient-boosted regression & classification
Pandas	2.1	Data manipulation and feature engineering
NumPy	1.26	Numerical computations
Plotly	5.20	Interactive charts (line, bar, pie, dual-axis)
Scikit-learn	1.4	Preprocessing, metrics, model selection
Pickle	built-in	Model serialisation and deserialisation

2.4 Library Description

XGBoost: Extreme Gradient Boosting library providing both regression (XGBRegressor) and classification (XGBClassifier) estimators with built-in early stopping, cross-validation, and feature importance extraction.

Pandas: DataFrame-based data manipulation library used for CSV ingestion, datetime parsing, rolling-window feature creation, groupby aggregations, and monthly trend analysis.

NumPy: Foundation library for vectorised numerical operations, array broadcasting, and efficient computation of error metrics.

Streamlit: Open-source Python framework that converts Python scripts into shareable web applications without requiring frontend coding. Used for all 4 dashboard pages.

Plotly: Interactive graphing library used for line charts (actual vs predicted), dual-axis charts (load + peak probability), histograms (error distribution), and donut charts (peak risk categories).

Scikit-learn: ML toolkit providing StandardScaler (feature normalisation), train_test_split, confusion_matrix, classification_report, and evaluation metrics.

Pickle: Python's native serialisation module used to save and reload trained model objects (all_results.pkl, predictions.pkl, peak_analysis.pkl, residuals.pkl).

CHAPTER 3: SYSTEM DESIGN

3.1 System Flowchart / Algorithm

The overall pipeline follows a sequential flow from raw data ingestion through feature engineering, dual-model training, evaluation, and finally deployment as a Streamlit application.

► START

- Load raw CSV: `power_system_load_cleaned.csv` (17,496 rows × 19 columns)
- Exploratory Data Analysis (EDA): check nulls, distributions, correlations
- Feature Engineering: temporal features (hour, DOW, month, season, week), lag features (`lag_1h`, `lag_1d`, `rolling_7d_avg`), price change, solar proxy, interaction terms
- Define target variables: `y_reg = load_demand_MW` | `y_cls = is_peak`
- Train-test split: 90 % train / 10 % test (time-ordered, no shuffle)
- Train XGBoost Regressor → evaluate: MAE, RMSE, MAPE, R^2
- Train XGBoost Classifier → evaluate: Precision, Recall, F1, AUC-ROC
- Serialise all results to .pkl files (`all_results`, `predictions`, `peak_analysis`, `residuals`)
- Launch Streamlit app → serve Dashboard, Forecast, Peak Risk, Performance pages
- **END**

3.2 Dataset Design

The dataset `power_system_load_cleaned.csv` contains 17,496 hourly records spanning January 2022 to December 2023 (2 full years). The table below describes all 19 columns.

Column Name	Data Type	Description
datetime	datetime	Hourly timestamp (YYYY-MM-DD HH:MM:SS)
load_demand_MW	float	Target: actual electricity load in Megawatts
temperature_C	float	Ambient temperature in Celsius
humidity_%	float	Relative humidity (%)
solar_irradiation_Wm2	float	Solar irradiance (W/m ²)
population	int	Regional population (socio-economic proxy)
gni_per_capita_usd	float	Gross National Income per capita (USD)
power_price_usd_per_kwh	float	Electricity price (USD/kWh)
day_of_week	int	0=Monday ... 6=Sunday
hour_of_day	int	Hour 0–23
is_holiday	int	Binary: 1 if public holiday
season	str	Spring / Summer / Autumn / Winter
year	int	2022 or 2023
month	int	1–12
week_of_year	int	ISO week number
lag_1_hour_load	float	Load value from 1 hour prior
lag_1_day_load	float	Load value from 24 hours prior
moving_avg_7_day	float	Rolling 7-day average load
hour	int	Duplicate of hour_of_day (used in Streamlit)

3.3 Preprocessing Steps

Null Handling: Verified zero null values in the cleaned dataset.

Datetime Parsing: Converted string datetime column to pandas datetime64 and set as index for time-series operations.

Peak Label Creation: Defined `is_peak = 1` when `load_demand_MW > 90th` percentile of the training set (≈ 340 MW) during high-load hours (22–23, 0–6).

Feature Scaling: Applied `StandardScaler` to continuous features before feeding into the classifier (XGBClassifier is not scale-invariant for probability calibration).

Lag Feature Creation: `lag_1_hour_load` and `lag_1_day_load` were shifted using pandas `.shift()` to prevent data leakage.

Rolling Window: 7-day rolling mean was computed on the training set, then propagated to test set using boundary values to avoid leakage.

Train-Test Split: Chronological 90/10 split: first 15,728 rows for training, last 1,768 rows (Dec 2023) for testing.

One-Hot Encoding: season column encoded using `pd.get_dummies()` with `drop_first=True`.

CHAPTER 4: DEVELOPMENT

4.1 Model Details & Source Code Overview

The development is organised in two files: model.ipynb (Jupyter Notebook for training and evaluation) and app.py (Streamlit web application for serving results). The training notebook produces four serialised pickle files that the web app loads at startup.

XGBoost Regressor — Key Hyperparameters:

- n_estimators: 500 | learning_rate: 0.05 | max_depth: 6
- subsample: 0.8 | colsample_bytree: 0.8 | reg_alpha: 0.1
- Early stopping: 50 rounds on validation MAE
- Train MAE: ~1.2 MW | Test MAE: 1.84 MW | Test R²: 0.9964

XGBoost Classifier — Key Hyperparameters:

- n_estimators: 300 | learning_rate: 0.1 | max_depth: 5
- scale_pos_weight = (non-peak count) / (peak count) ≈ 9 (handles class imbalance)
- Test Precision: 0.889 | Test Recall: 1.000 | Test F1: 0.941 | AUC-ROC: 0.999
- Confusion Matrix: TP=8, FP=1, TN=159, FN=0 (zero missed peak hours)

4.2 Application UI Design (Streamlit)

The app.py Streamlit application is structured into four pages accessible via a sidebar radio selector.

Dashboard: Displays 4 KPI cards (MAE, RMSE, F1, Peak %), a 7-day actual-vs-predicted chart, a peak risk donut chart, and a 24-hour ahead dual-axis forecast panel.

Forecast Analysis: Lets the operator choose a time window (24–720 hours) and chart type. Shows error distribution histogram and error-vs-predicted scatter plot.

Peak Risk Detection: Shows precision/recall/F1 metrics, a slider-controlled peak probability timeline overlaid with actual load, and risk threshold lines at 30 % and 70 %.

Model Performance: Displays all stored evaluation metrics from all_results.pkl and a bar chart of feature correlations with load demand.

4.3 Test Reports

The following table summarises the formal evaluation results on the December 2023 hold-out test set (168 hourly records).

Model	Metric	Value
Load Forecaster	MAE	1.84 MW
Load Forecaster	RMSE	2.38 MW
Load Forecaster	MAPE	0.68 %
Load Forecaster	MPE	-0.12 %
Load Forecaster	R ²	0.9964
Load Forecaster	Max Error	7.99 MW
Peak Detector	Precision	0.889
Peak Detector	Recall	1.000
Peak Detector	F1-Score	0.941
Peak Detector	AUC-ROC	0.999
Peak Detector	Confusion Matrix	TP=8, FP=1, TN=159, FN=0

CHAPTER 5: PROPOSED ENHANCEMENTS

Real-Time SCADA Integration: Connect the pipeline to live SCADA feeds so the model receives actual sensor data every hour and updates the 24-hour forecast automatically.

LSTM / Transformer Models: Explore sequence-to-sequence deep-learning models (LSTM, Temporal Fusion Transformer) to capture longer-range temporal dependencies beyond the current lag features.

Probabilistic Forecasting: Implement quantile regression or conformal prediction intervals to provide uncertainty bounds around forecasts, enabling risk-aware dispatch decisions.

Longer Training Window: Incorporate 5–10 years of historical data to better capture seasonal drift, long-term growth trends, and rare demand events (e.g., heat-wave spikes).

Mobile-Responsive UI: Upgrade the Streamlit app to a production-grade React or Angular front-end with role-based access control for different utility stakeholders.

Anomaly Detection Module: Add an unsupervised anomaly detector (Isolation Forest or Autoencoder) to flag data-quality issues in incoming sensor streams before they corrupt forecasts.

Carbon Emission Overlay: Integrate regional carbon intensity data to overlay CO₂ emission estimates alongside load forecasts, supporting net-zero planning.

CHAPTER 6: CONCLUSION

This project successfully demonstrates that machine learning — specifically Extreme Gradient Boosting — can deliver highly accurate electricity demand forecasting and peak-risk detection from a carefully engineered feature set. The dual-model pipeline achieves an R^2 of 0.9964 for regression and an AUC-ROC of 0.999 for peak classification on an unseen December 2023 test set, confirming strong generalisation with virtually no missed peak events (Recall = 1.0).

Peak demand hours ($\approx 10\%$ of all hours) are heavily concentrated in late-night and early-morning windows (22:00–06:00) and in spring/summer months, driven by residential heating and cooling loads. This seasonal and diurnal insight enables targeted generation scheduling that can prevent grid stress.

The accompanying Streamlit web application bridges the gap between data science and grid operations, providing non-technical users with intuitive visualisations, 24-hour ahead demand schedules, and colour-coded risk alerts — all within a zero-installation browser interface.

Future work should focus on live SCADA integration, deep-learning sequence models, probabilistic forecast intervals, and a carbon-aware dispatch layer to support India's net-zero ambitions in the energy sector.

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